

## Maze Madness Extension 5 — Redstone AND Solver

**Topic:** Redstone AND Rules + Single Checks · **Course:** Maze Madness · **Level:** Extension (Intermediate) · **Time:** about 45 minutes

Same **Lesson 3 world**. Find the **hardest maze** your teacher shows a picture of — tiles with **two** redstones need an **AND** (both true).

### The rules you need

- RS **left** → turn right · RS **right** → turn left (single)
- RS **below** → move up 1 (single)
- **#1** RS left **and** below → up 1, turn right (AND)
- **#2** RS right **and** below → up 1, turn left (AND)
- **#4** RS left **and** right → up 5 (AND)

Chat: **l** turn left · **r** turn right · **run** start · **rl** bring the agent back.

### 1 · Read the Chat Commands

A chat word binds to an action. Type the word, the agent runs the code.

```
def turn_left():
    agent.turn(LEFT_TURN)
player.on_chat("l", turn_left)
```

You type **l**. What does the agent do?

```
def go_back():
    agent.teleport_to_player()
player.on_chat("rl", go_back)
```

You type r1 . Where does the agent go? When is that handy?

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## 2 · Trace the Solver 🔍

This loop runs when you type `run`.

```
while True:
    if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, RIGHT):
        agent.move(UP, 5)
    elif agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, DOWN):
        agent.move(UP, 1)
        agent.turn(RIGHT_TURN)
    agent.move(FORWARD, 1)
```

Rule #4 is `RS LEFT and RS RIGHT`. Which line runs there, and how far does the agent climb?

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Rule #1 is `RS LEFT and RS DOWN`. Which line runs, and what two things does it do?

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A tile has redstone on the left only — nothing below, nothing on the right. Does either `if` run? Why not?

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### 3 · Fill the Blanks

Rule #2: move up 1 and turn left **only when both are true** — redstone right and redstone below.

```
if agent.detect(REDSTONE, _____) _____ agent.detect(REDSTONE, DOWN):  
    agent.move(UP, 1)  
    agent.turn(_____)
```

Word bank: RIGHT · and · LEFT\_TURN

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## 4 · Spot the Bug 🐛

**Bug A** — Rule #1 should fire **only** when redstone is left **and** below. It checks redstone left only, so it fires on a plain left marker with nothing below it.

```
if agent.detect(REDSTONE, LEFT):  
    agent.move(UP, 1)  
    agent.turn(RIGHT_TURN)  
    agent.move(FORWARD, 1)
```

**Hint:** add the second check with `and` .

**Write the fix:**

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**One line: why was it wrong?**

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## 5 · Show Your Work 📷 🎥

Open the Lesson 3 world, find the hardest maze from the picture, and run your solver until the agent reaches the **diamond block** goal.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

**Write your lines here, then say them in your video.**

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Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.